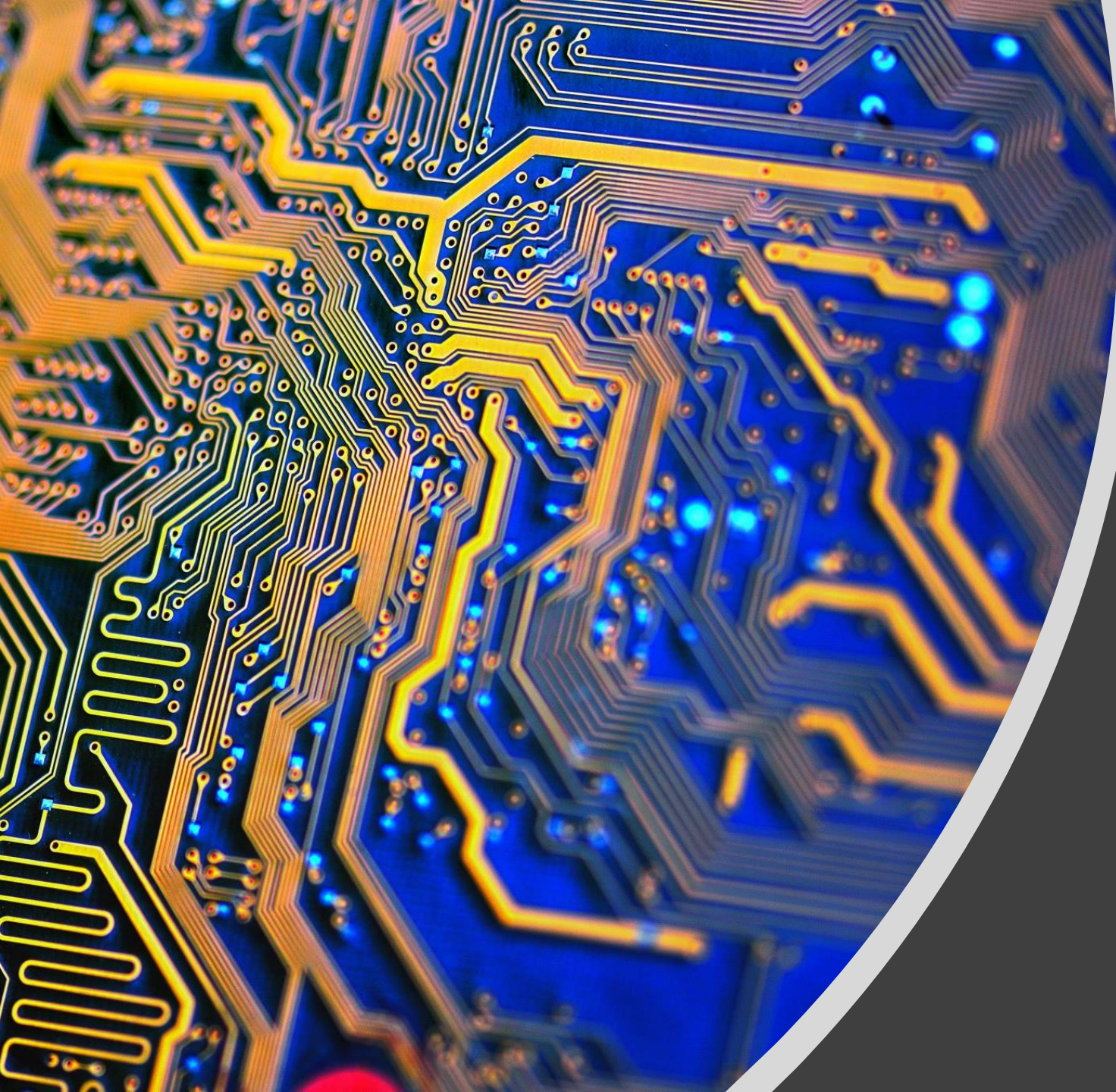




CS683: Computer Architecture?

Computer Architecture 2050 (seriously?)



Quantum Computer

Myths

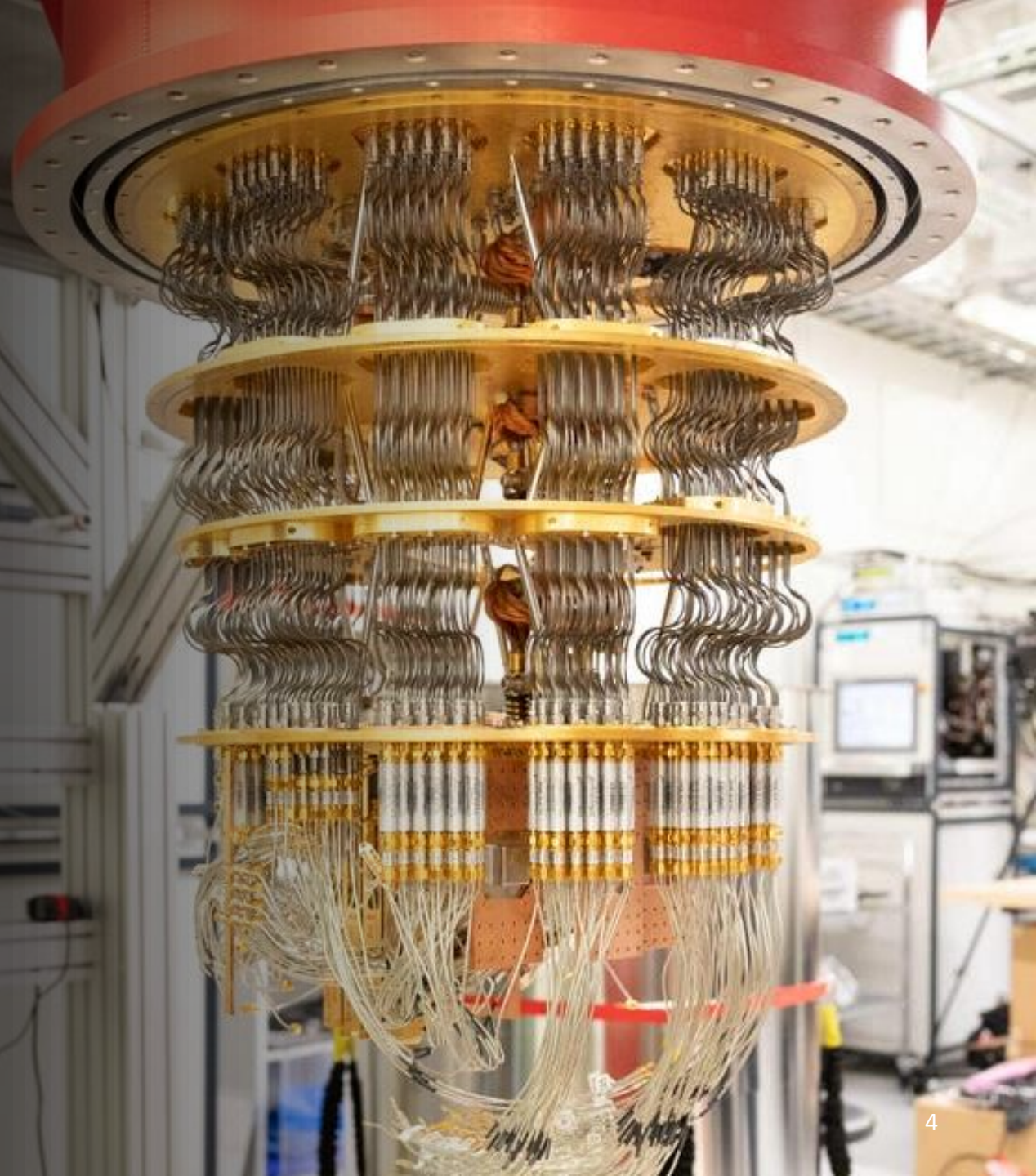
Fastest computer: NO

One quantum computer: 100 supercomputers, NO

One quantum computer: 100 datacenters, NO

What is it then? It is a quantum computer 😊

The quantum computer





Bits to qbits

Bit: 0 or 1

Qubit: values between 0 to 1, huge quantum space (0.20 0 and 0.80 1 can be one state)

Utility?



Factorization

Question: Find the factors of a 4000-bit number

Use all the super-computers of the entire world, use all the energy of the world

It will take more than 1000 years ☹️ ☹️ ☹️

Quantum paradigm can do it in hours.

Is there a real
need/application?

Factorization is a big deal !!

If not, then what about

problems like how cells/organisms
behave

Definitely not to speedup matrix
multiplications



Moving On





Arch: 2050

Vertically integrated arch.

Domain specific

Compilers, runtimes, architecture

All tied together

Hw-sw contract revisited, data-centric

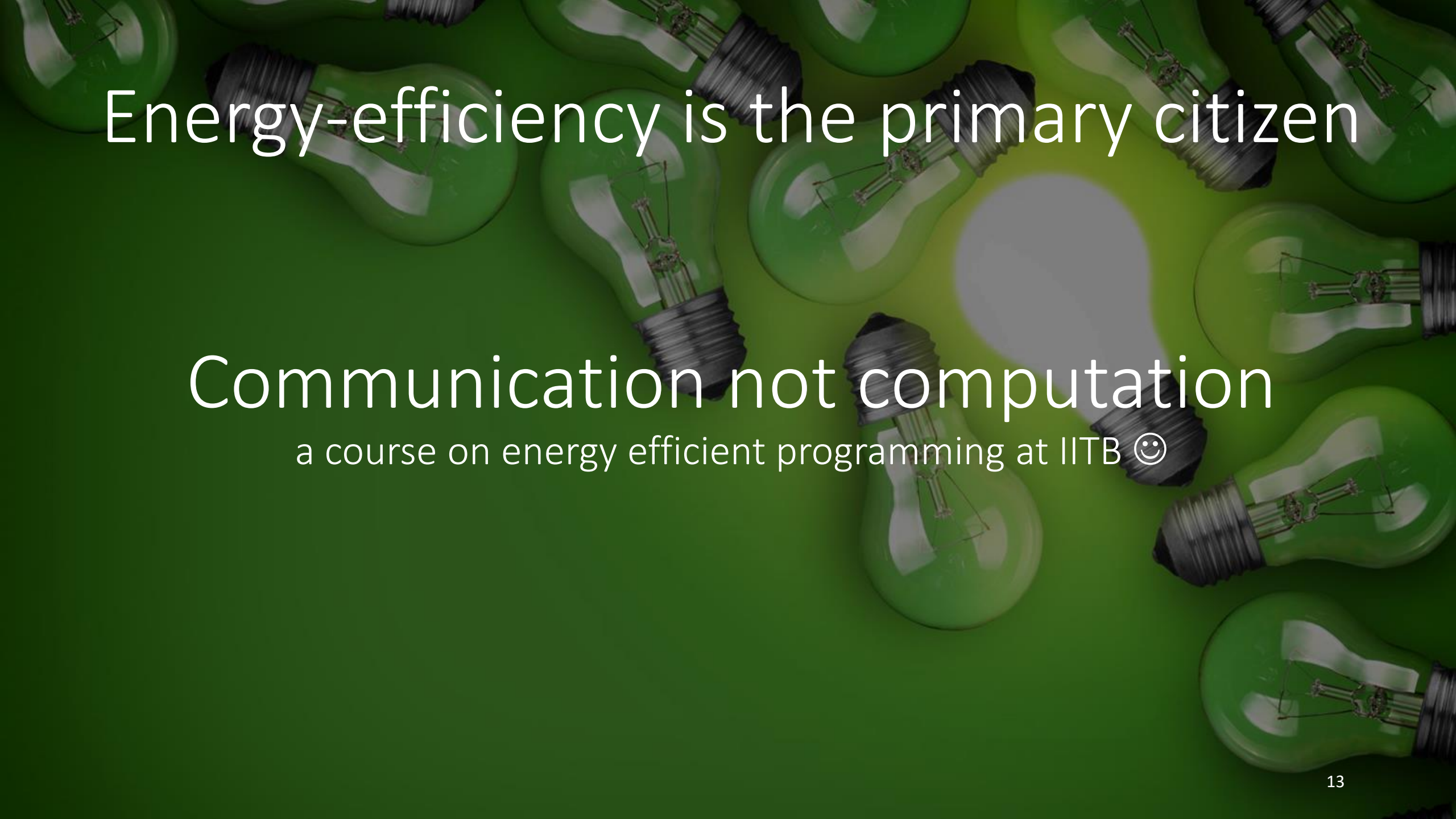
Applications to look for atleast till 2030



Machine Learning



IoT (N billion devices by 2025)



Energy-efficiency is the primary citizen

Communication not computation

a course on energy efficient programming at IITB 😊



Carbon Emission (sustainable architecture)



WALMART AIMS FOR ZERO GREENHOUSE GAS EMISSIONS BY 2040

Axios, Sept 2020

Google, Facebook, Microsoft to Go Carbon-Free by 2030

Science Times, Sept 2020

Inside Apple's green revolution: Can it make a carbon neutral iPhone?

British GQ, Sept 2020

Security: Redesign microarchitecture (CS773 is waiting for you)



Nah, redesign system stack



Clearly specify who can do what



Simple protocol: If I cannot see,
touch some I should not have
any information about some

The other side

Industry is pushing for more automated stuff

Good for end users 😊 😊

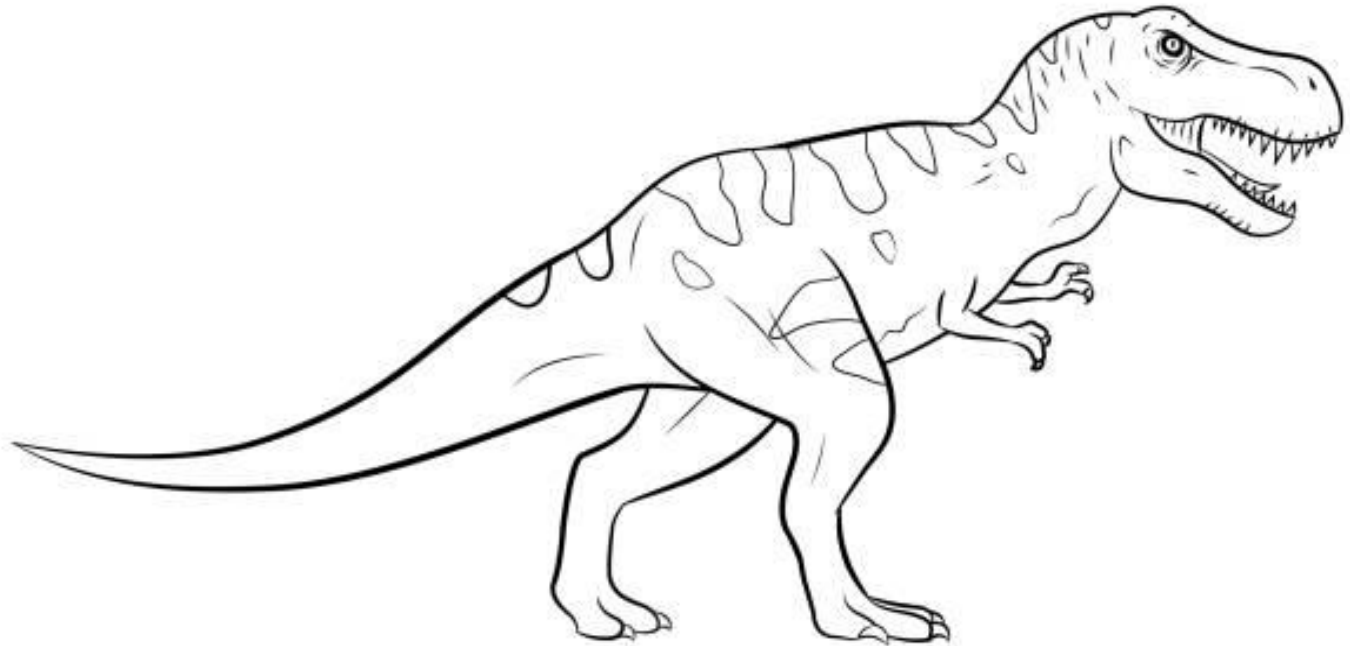
Bad for CS students (especially systems folks) 😞 😞

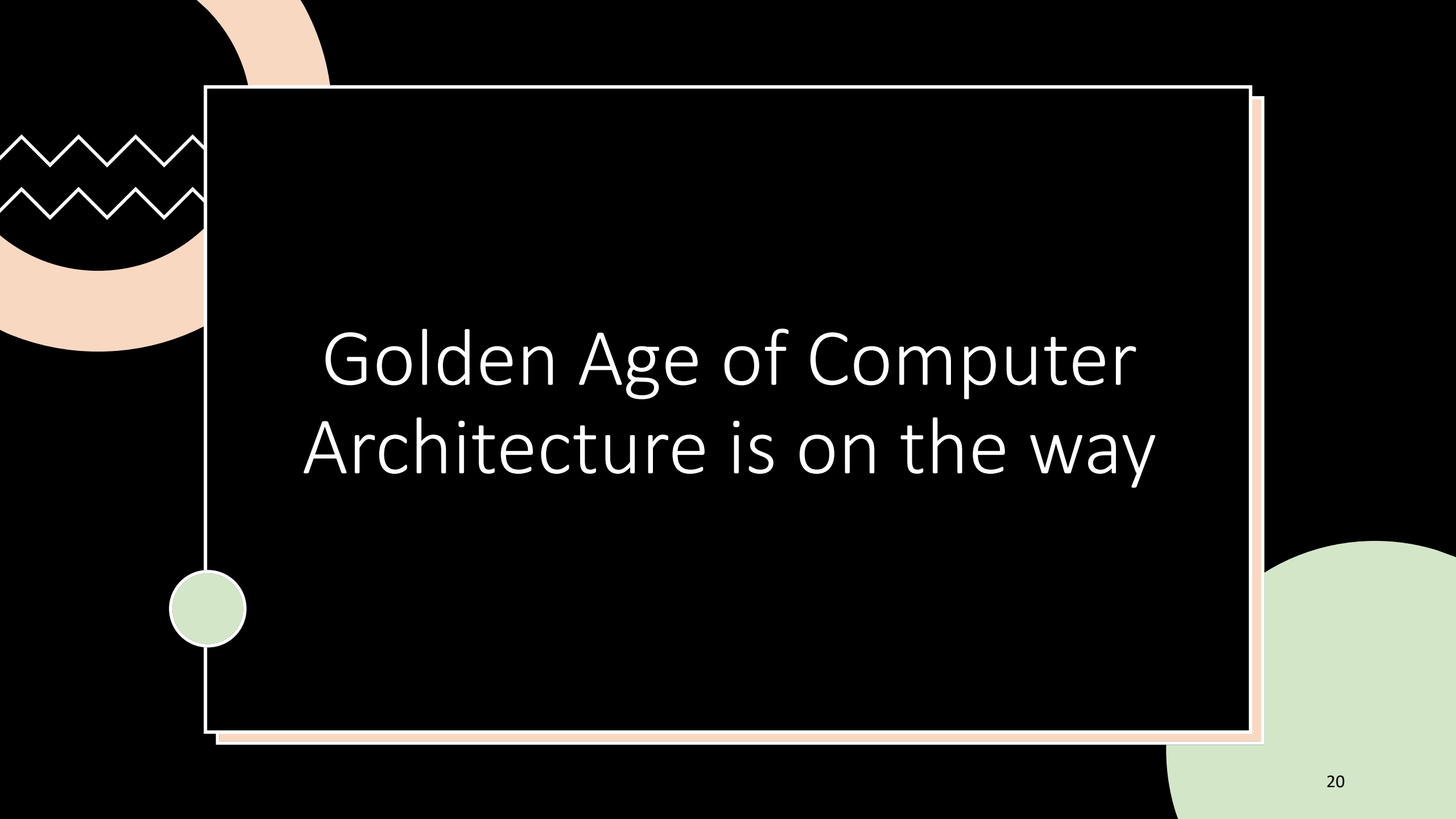
By 2050, we may end up with a critical mass of What?



You heard it
right, no I am
just kidding

CS683 Instructor in 2050? Nah





Golden Age of Computer
Architecture is on the way